# Digit Concept - Digits extraction

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$7789 \% 10 = 9$ ?

$$\begin{array}{r|l} 7789 & 10 \\ \hline -7780 & 778 \\ \hline \boxed{9} & \end{array}$$

mod returns the reimender of a division

next step

/10

How to get the next digit 8?

$778 \% 10 = \boxed{8}$

/10

$77 \% 10 = \boxed{7}$

/10

$7 \% 10 = \boxed{7}$

/10

0 stop $\Longrightarrow$ we extracted all the digits in reverse

## Pseudo Code:

```
N: the number
while( n > 0){
    lastDigit = n % 10
    n = n/10
}
```

# Digit Concept - Digits count

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## Count the number of digits

method 1

```
count := 0
while( n > 0){
    count++
    n = n/10
}
```

number of digits = the number of division by 10

$= \log_{10}(n) + 1$

$= \log_{10}(7789) + 1$

$= 3.89 + 1 = 4,89$

$\Rightarrow$ 4 digits TC: $O(\log_{10}(n))$

method 2

```
int Count( int n){
    return (int)( log10(n) + 1 )
}
```

TC: $O(1)$

# Digit Concept - Reverse & Palindrome Number

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## Examples:

123 → 321 / 120 → 21

```
int getReverseNumber(int n){
    int result = 0
    while (n>0) {
        lastDigit = n % 10
        result = result x 10 + LD
        n = n / 10
    }
    return result}
```

0 ×10
0 +LD
3 ×10
30 +LD
32 ×10
320 +LD
321

## Palindrome number

It's a number that reads the same as backward as forward, examples: 121, 1331, 4554

```
bool isPalindrome(int n){
    int copy = n
    pn = getReverseNumber(n)
    if(pn == copy){
        return true
    }
    return false
}
```

# Digit Concept - Armstrong Number

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It's a number that is equal to the sum of its digits each raised to the power of the number of digits. example: $153 = 1^3 + 5^3 + 3^3$

Solution:

Loop over digits and calculate the sum of the digit power l

```
bool isArmstrongNumber( int n){
    int l = to_string(n).length()
    int nCopy = n
    int sum = 0
    while(n>0){
        sum = pow(n%10, l)
        n = n/10
    }
    if (nCopy == sum){
        return true
    }
    return false
}
```

# Digit Concept - get all Divisors

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